



Homework 1 report

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Preprocessing

Prior to fitting, GDP per capita and population independent variables were log transformed due to them being right skewed. Then their distributions were much closer to a normal PDF, which is what a linear regression expects. All of the numerical independent variables (everything but continent) were also z score standardized (mean = 0, sd = 1). Lastly, the continent was transformed into 4 dummy variables. Its every distinct value was a new column, except Africa, which was the reference one and so it was automatically omitted.

Answers (concise, with quantified certainty)

1. Is life expectancy improving over time?

Yes. The posterior mean for the standardized-year slope is **4.07 years per 1 SD increase $\pm$ 0.002**. All posterior draws are **positive**, (with posterior probability 1.00 ± 0.00), so we are highly confident that life expectancy has increased over time in this dataset.

2. How are GDP per capita and population correlated with life expectancy?

Conditional (controlled) effects from the model:

- A 1 SD increase in *log GDP per capita* is associated with **6.39 years $\pm$ 0.0048** increase in life expectancy.
- A 1 SD increase in *log population* is associated with **0.28 years $\pm$ 0.0025** increase.

Raw (Pearson) correlations: $\text{cor}(\log \text{GDP}, \text{life}) = 0.8076$, $\text{cor}(\log \text{pop}, \text{life}) = 0.1884$. These raw correlations show a strong positive association for GDP and a weak positive association for population.

3. Expected life for you (born 2002, Europe; average GDP/pop)

Posterior predictive estimate for the conditional mean life expectancy (Europe, birth year 2002, $\log \text{GDP} = 0$, $\log \text{pop} = 0$) is

71.02 $\pm$ 0.0078 years

4. Probability that an *average* European (2002) lives longer than average counterparts

This model compares a person living in Europe and born 2002 with all other parameters set to their mean (which is the 0, because of z score normalization).

$$P(\mu_{\text{Europe}} > \mu_{\text{others}}) \approx \mathbf{0.90 \pm 0.0071}.$$

5. Probability a random individual European (2002) outlives a random individual elsewhere

We compare similarly as before, just that now we add the normally distributed noise of the target variable, which we get from draws of the sigma parameter. We get:

$$P(Y_{\text{Europe}} > Y_{\text{others}}) \approx \mathbf{0.71 \pm 0.0052}.$$

This probability is lower than the mean-vs-mean value because individual-level residual variation (here $\sigma \approx 5.8$ years) increases overlap between outcome distributions.

6. How much longer than your professor (born 1985)?

Assuming both in Europe (avg. GDP/pop):

- Posterior mean difference (you – professor): $\approx$ **4.01 years $\pm$ 0.0024**.
- 95% credible interval for the difference: [3.71, 4.31] years.
- Because the entire 95% interval is positive, the posterior probability that you outlive the professor (random individuals, ignoring residual sampling here) is effectively ≈ 1.0 under the model's assumptions.

Discussion & limitations (brief)

Main limitations & assumptions:

1. **Linear model form and homoskedastic residuals:** the model assumes linear effects (on the standardized predictors) and constant residual variance. If relationships are nonlinear or heteroskedastic, estimates may be biased.